# Residual Surrogates for Nonlinear DSGE Estimation: An Auditable Target-Corrected Workflow

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**Abstract**

Repeated nonlinear solution can dominate likelihood-based estimation of dynamic stochastic general equilibrium models. We develop an auditable workflow in which a neural network learns the difference between nonlinear extended-path trajectories and a parameter-specific first-order solution. Cheap surrogate gradients form Hamiltonian proposals; an endpoint Metropolis test preserves the density evaluated there, which need not be the structural nonlinear posterior. On the complete 23-equation Galí model, a clean order-one, three-node stochastic preflight solves all eight prespecified path evaluations, spanning seven unique policy-rule points, at a five-period terminal horizon. The largest equation residual is $1.93 \times 10^{-10}$. This result establishes a reduced numerical pilot, not horizon convergence: the hard design fails under both solvers at horizon six, and first-step outcomes still move between horizons four and five. Separate clean-revision comparisons against MacroModelling.jl match the non-stochastic steady state and first-order Schur solution and, at a three-period horizon, the complete order-one sparse nonlinear scenario tree. Across eight evaluations, the worst tree and expected-path discrepancies are $1.75 \times 10^{-7}$ and $7.22 \times 10^{-9}$. These are core and reduced-horizon trajectory checks, not production-horizon likelihood or posterior parity. We therefore pair the method with executable gates for stochastic-solution accuracy, independent parity, untouched test performance, posterior agreement, multichain sampling, and same-work timing.

## 1 Introduction

Likelihood-based estimation of a nonlinear DSGE model repeats a costly numerical experiment at every parameter proposal. A single evaluation may require a steady state, a local solution, nonlinear conditional trajectories, latent-shock inversion, and measurement-error evaluation. Posterior simulation then repeats this stack thousands of times. Approximation is attractive, but an unmeasured approximation can change the objective, the posterior geometry, and even the numerical support of the model.

This paper develops a residual-surrogate workflow designed around that risk. The method solves the complete nonlinear equations at designed parameter, state, and shock inputs, then learns the difference between those solutions and a parameter-specific first-order reduced-order model (ROM). The ROM supplies a disciplined local baseline; the network is responsible only for the remaining nonlinearity. During sampling, the cheap differentiable objective supplies the Hamiltonian force. A separately declared density is evaluated at each proposal endpoint. Standard Metropolis correction then preserves that declared density when the numerical proposal map is reversible and volume preserving.

The preserved Galí pilots establish that this software path composes on the complete 23-equation nonlinear model while only the estimated parameter vector is reduced. Across 80 deterministic

extended-path targets, the largest recorded equation residual is $9.92 \times 10^{-9}$. Python and JAX evaluations of the same surrogate profile objective differ by at most $1.71 \times 10^{-13}$ at the audited calibration. A calibration block shows a 53.9 percent mean reduction in state error relative to the ROM after per-output shrinkage is fitted on that same block. The last number is not an untouched test result, the HMC budgets are not convergence evidence, and no preserved run yet establishes a nonlinear posterior speed advantage.

The first stochastic positive-order pilot is deliberately retained despite failing its hard gate. Only three of eight parameter paths completed a ten-period, order-one, three-node calculation, while five paths failed at the initial simulated period. A later clean preflight isolates the solver from training and inference. All eight prespecified path evaluations, spanning seven unique policy-rule points because the reference is evaluated twice, pass the $10^{-8}$ residual gate at horizons three through five, but the hard design fails under both sequential and batched solvers at horizon six. Moreover, the horizon-four- to-five change reaches $3.96 \times 10^{-4}$ for log output, $1.22 \times 10^{-3}$ for annualized inflation, and $1.39 \times 10^{-3}$ for the annualized policy rate. Residual convergence at horizon five therefore does not establish terminal-horizon convergence.

Independent cross-language comparisons close two scoped implementation risks. The Python and MacroModelling.jl calculations use byte-identical copies of the complete Galí source, inject the policy coefficients by name despite their different internal parameter orders, and retain the duplicated reference in the eight-case denominator. Every non-stochastic steady state and first-order Schur solution passes its frozen tolerance. A second comparison activates all three shock channels and matches the complete three-period, order-one sparse nonlinear scenario tree, rather than only its central path. This reduced-horizon result does not establish production-horizon trajectory or likelihood parity.

The paper makes three contributions. First, it separates three objects often conflated in surrogate estimation: nonlinear trajectories used as labels, a surrogate profile objective used for cheap differentiation, and the nonlinear likelihood required for a structural-posterior claim. Second, it integrates parameter-dependent DSGE solution, residual learning, bounded-parameter HMC, and explicit failure handling in a content-addressed workflow. Third, it turns publication claims into executable gates covering stochastic expectations, cross-language parity, untouched test accuracy, posterior fidelity, sampler diagnostics, and full computational accounting.

The paper does not claim that endpoint correction or neural solution methods are new by themselves. Its narrow claim is that nonlinear DSGE surrogates can be made auditable enough to determine when they are safe and useful for estimation. The rest of the paper defines the estimands, describes the numerical workflow, proves the target-invariance statement used by the sampler, and reports the preserved engineering evidence before stating the confirmatory design and stopping rules.

### 1.1 Relation to the literature

The estimation problem sits between nonlinear DSGE solution, neural approximation, and accelerated MCMC. Extended path provides the nonlinear trajectory construction [Fair and Taylor, 1983]; stochastic extended path replaces future-shock means with numerical integration over conditional expectations [Adjemian and Juillard, 2026]. Likelihood construction for nonlinear DSGE models remains model- and solution-dependent [Kollmann, 2017], which is why this paper treats the trajectory solver, latent-shock treatment, and statistical objective as separate validation targets.

Neural networks have been used to solve high-dimensional dynamic economic models [Maliar et al., 2021]. Here the network does not replace the economic model wholesale: it learns the full-minus-first-order discrepancy and is always benchmarked against the ROM. Surrogate-gradient

HMC and neural-gradient HMC already establish the proposal-side idea [Zhang et al., 2017, Li et al., 2019]. Our focus is the less forgiving DSGE case in which the endpoint density may contain parameter-dependent solution failures and may differ from the density used to generate the force. The independent implementation benchmark uses MacroModelling.jl [Kockerols, 2023], while the JAX path builds on the differentiable state-space perspective of Childers et al. [2022].

## 2 Economic model and statistical objects

Let $\theta \in \Theta$ denote structural parameters, $s_t$ predetermined states, $x_t$ endogenous variables, and $\varepsilon_t$ structural shocks. Write the nonlinear equilibrium system as

$$F(x_{t-1}, x_t, \mathbb{E}_t x_{t+1}, \varepsilon_t; \theta) = 0. \tag{1}$$

For a given parameter vector and shock design, a nonlinear extended-path solver produces a trajectory $x^F_{1:T}(\theta)$ from the full-order model (FOM). A first-order solution produces the cheaper trajectory $x^R_{1:T}(\theta)$. We fit a network $r_\psi$ to the residual

$$\Delta x_t(\theta) = x^F_t(\theta) - x^R_t(\theta), \qquad \widehat{x}^F_t(\theta) = x^R_t(\theta) + r_\psi(z_t, \theta), \tag{2}$$

where $z_t$ contains the states and shocks available at the prediction point. Parameter vectors, rather than individual rows, are assigned to data partitions. This prevents periods from the same parameter-specific trajectory from appearing on both sides of a split.

The target nonlinear posterior for data $y_{1:T}$ is

$$\pi_F(\theta \mid y) \propto p(\theta) L_F(y_{1:T} \mid \theta), \tag{3}$$

where $L_F$ denotes an independently validated nonlinear likelihood with a declared treatment of latent shocks and every required change-of-variables term. The present implementation instead defines a surrogate inversion objective $\widetilde{L}_\psi$ by replacing nonlinear trajectories with (2), optimizing latent shocks, and evaluating the shock and measurement penalties at that optimizer. Unless an integration, Laplace adjustment, or required change-of-variables term is supplied, this is a profile or pseudo-likelihood objective rather than a marginal likelihood. The two objects are therefore scientifically distinct. Agreement between Python and JAX evaluations of $\widetilde{L}_\psi$ is implementation parity; it is not evidence that $\widetilde{L}_\psi = L_F$.

More explicitly, let $h_\psi(s_t, \varepsilon_t, \theta)$ map a state and structural shock into the observables implied by the ROM plus residual surrogate. At each date, the implementation computes

$$\widehat{\varepsilon}_t(\theta) = \arg\min_{\varepsilon} \left\{ \|R^{-1/2}[y_t - h_\psi(s_t, \varepsilon, \theta)]\|_2^2 + \|Q^{-1/2}\varepsilon\|_2^2 \right\}, \tag{4}$$

using a damped Gauss–Newton iteration, and then sums the Gaussian penalties evaluated at $\widehat{\varepsilon}_t(\theta)$. Here $R$ and $Q$ are diagonal in the pilot. The state passed to date $t+1$ follows the configured ROM or surrogate replay. Equation (4) makes the missing operation clear: the latent shocks are optimized, not integrated out. It also clarifies why agreement between two implementations of this calculation is a software check rather than likelihood validation.

Table 1 records the three numerical objects used below. A trajectory label is not a density, and a profile objective is not a marginal likelihood. Keeping their names distinct prevents a correct MCMC kernel from being used to support the wrong inferential claim.

The benchmark is the complete nonlinear New Keynesian system distributed with the MacroModelling.jl examples [Kockerols, 2023] and associated with Galí [2015]. It contains 23 equations, 16

Table 1: Numerical objects and the claims they can support

| Object | Construction | Valid interpretation |
|---|---|---|
| Nonlinear label $x^F$ | Extended-path solution of the complete equilibrium system | Supervised target for trajectory approximation; not a probability density |
| Surrogate objective $\widetilde{L}_\psi$ | ROM plus learned residual, with shocks profiled by (4) | A declared pseudo-posterior target when combined with a prior and transform Jacobian |
| Nonlinear likelihood $L_F$ | Independently validated statistical likelihood with latent variables and Jacobians treated explicitly | Structural likelihood and posterior in (3) |

declared parameters, and three structural shocks. The reduced experiment estimates the Taylor-rule responses to inflation and output, $\phi_\pi$ and $\phi_y$, while retaining every model equation. Reducing the parameter dimension makes a numerical posterior reference feasible; it does not replace the model with the familiar three-equation linearization.

### 2.1 Pilot data and parameter design

The preserved software fixture contains ten synthetic observations of log output, annualized inflation, and the annualized nominal interest rate. The measurement standard deviation is 0.01 for each observable. The three declared structural shocks are technology, monetary-policy, and preference shocks, with model innovation scales 0.01, 0.0025, and 0.05 multiplying unit-variance innovations inside the equilibrium equations. These observations are useful for testing the inversion and sampling stack; they are not an empirical data application. The confirmatory experiment freezes a documented data-generating parameter, simulation seed, burn-in, initial condition, and shock path before training or posterior evaluation.

The deterministic default pilot contains eight parameter vectors and ten periods per vector. The reference calibration is included, and the two estimated policy coefficients are perturbed by the fixed design while the remaining 14 parameters stay at their declared calibration. Two complete parameter groups form the calibration block. The positive-order pilot uses the same reduced parameter design so that its purpose remains solver feasibility rather than a claim about broad parameter-space coverage.

## 3 Residual-surrogate construction

For training case $i$, let $u_i = (z_i, \theta_i)$ and $d_i = x_i^F - x_i^R$. Input and output location and scale are estimated only from the training partition. In standardized coordinates, the implemented data-fit loss is

$$\widehat{\psi} = \arg\min_{\psi} \frac{1}{\sum_{i\in\mathcal{I}_{\mathrm{tr}}} w_i} \sum_{i\in\mathcal{I}_{\mathrm{tr}}} w_i \left\| \widetilde{d}_i - r_\psi(\widetilde{u}_i) \right\|_2^2, \tag{5}$$

with zero weight for masked failures. The optimizer is AdamW with gradient clipping. The network embeds states and shocks, conditions hidden activations on the full estimated parameter vector through feature-wise affine modulation, and uses residual blocks with SiLU activations. Deployment restores the original units and adds the predicted discrepancy to the ROM.

The preserved default configuration uses one residual block, 64 hidden units, 400 epochs, batches of 32, and 64-bit training arithmetic. These choices are engineering settings, not selected

hyperparameters. Confirmatory learning curves vary target count and network capacity using only training and calibration data; the untouched test partition is evaluated once for the frozen specification.

The implementation can fit an additional per-output shrinkage coefficient in $[0, 1]$ after network training. In the preserved pilots, those coefficients were fitted on the same parameter block used to report the resulting RMSE. That block is therefore a calibration set, not an untouched test set. We retain the values as engineering diagnostics but do not treat them as out-of-sample evidence. Publication launchers disable this calibration; a future use of shrinkage requires a distinct calibration partition and an untouched test partition.

### 3.1 Offline and online computations

The offline stage draws complete parameter vectors, solves a parameter-specific steady state and first-order ROM, and generates nonlinear extended-path targets conditional on designed state and shock paths. Each attempted target retains a case identifier, solver status, residual norm, and parameter-group identifier. Only the training partition determines feature scaling and network weights. Validation partitions are disjoint in complete parameter vectors, not merely in individual rows. The residual network and every preprocessing constant are then frozen and serialized with the model identity and source hashes.

The online stage transforms an unconstrained vector $q$ to bounded structural parameters, adds the transform Jacobian to the declared log target, rebuilds the parameter-dependent objects required by the selected runtime, and evaluates either the profile objective or a nonlinear reference. A surrogate-gradient trajectory costs many cheap force evaluations but only one new target evaluation per transition. This accounting—rather than draws per second alone—is central when the endpoint target contains a nonlinear solution.

Extended path traces back to Fair and Taylor [1983]. Deterministic extended path sets future innovations to their conditional means and therefore does not capture Jensen effects in nonlinear expectations. Stochastic extended path approximates those conditional expectations over future shocks [Adjemian and Juillard, 2026]. This is why positive branching order, multiple nodes, and active variation in every shock are hard requirements for confirmatory evidence, even though order zero is useful for debugging deterministic nonlinearity.

## 4 Target-corrected surrogate trajectories

Let $q$ be an unconstrained parameterization of $\theta$, let $p$ be Gaussian momentum with kinetic energy $K(p)$, and let $\pi_T(q)$ be the density the sampler is intended to preserve. A cheaper differentiable density $\pi_P(q)$ defines the force used in $L$ leapfrog steps. Denote the resulting deterministic map by $(q', p') = \Phi_{\pi_P}(q, p)$. The acceptance probability is

$$\alpha\{(q,p),(q',p')\} = \min\left\{1, \exp\left[-H_T(q',p') + H_T(q,p)\right]\right\}, \quad H_T(q,p) = -\log \pi_T(q) + K(p). \tag{6}$$

**Proposition 1** (Target invariance)**.** *Suppose $\Phi_{\pi_P}$ is reversible after momentum negation and volume preserving, and $\pi_T$ is evaluated at both endpoints used in* (6)*. Then the Metropolis transition leaves $\pi_T$ invariant, even when $\pi_P \neq \pi_T$.*

*Proof.* On the extended state $z = (q, p)$, the target density is proportional to $\exp\{-H_T(z)\}$. Compose the leapfrog map with momentum negation. By the assumptions, the resulting map is an involution with unit Jacobian determinant. The deterministic Metropolis ratio is therefore $\exp\{-H_T(z') + H_T(z)\}$, and (6) gives detailed balance on the extended state. Marginalizing the

refreshed momentum leaves $\pi_T(q)$ invariant. The construction of $\Phi_{\pi_P}$ affects acceptance but not this invariance argument. □

One transition has four auditable steps. Draw fresh Gaussian momentum; apply $L$ leapfrog steps using only $\nabla \log \pi_P$; reject immediately if the proposal or target is non-finite; otherwise apply (6) using $\pi_T$ at the two endpoints. Step size and mass-matrix adaptation end with warmup. The frozen network, normalization constants, parameter transform, and target implementation are part of the recorded sampling state.

The proposition is the standard deterministic-proposal Metropolis argument for leapfrog HMC [Neal, 2011]. It places two practical restrictions on the implementation. First, the surrogate and every preprocessing transformation used in the force must be frozen during production sampling. Second, "exact" always modifies the declared target. If $\pi_T \propto p(\theta)\widetilde{L}_\psi$, the chain is exact for the surrogate-defined posterior, not for (3). Exact nonlinear posterior claims require endpoint evaluation of $L_F$ itself.

For bounded parameters $\theta_j \in (a_j, b_j)$, the implementation samples $q_j \in \mathbb{R}$ and uses

$$\theta_j = a_j + (b_j - a_j)\,\mathrm{logit}^{-1}(q_j), \qquad \log J(q) = \sum_j \left[\log(b_j - a_j) + \log \sigma(q_j) + \log\{1 - \sigma(q_j)\}\right]. \quad (7)$$

Thus the endpoint target is defined in $q$ coordinates and must include $\log J(q)$. Invalid steady states, indeterminate local solutions, failed nonlinear paths, or non-finite objective values must be rejected as points outside support. A large finite penalty is not equivalent: it creates an artificial plateau with positive density. The preserved pilot still uses such a sentinel and therefore cannot support an exact structural-posterior claim.

The sampler has three implementation-level regression checks. When proposal and target densities coincide, its draws and accept/reject decisions match the ordinary static-HMC implementation under the same key. Under a deliberately shifted and rescaled proposal force, a fixed-seed multi-chain test recovers the known moments of a standard-normal target after endpoint correction. Finally, non-finite candidate targets receive zero acceptance probability. These tests check kernel plumbing; they do not replace convergence and posterior-fidelity evidence for the DSGE application.

This distinction is especially relevant for differentiable state-space and DSGE estimation, where automatic differentiation improves computation but does not resolve approximation error in the target [Childers et al., 2022]. Related work has made nonlinear DSGE likelihoods more tractable [Kollmann, 2017]; our evaluation therefore treats accuracy against an independently computed nonlinear reference as a primary outcome.

## 5 Validation design

The confirmatory design has two tiers. Tier 1 estimates $(\phi_\pi, \phi_y)$ and uses a dense posterior grid. Grid resolution is doubled until posterior moments and interval endpoints stabilize. This tier isolates numerical correctness from high-dimensional MCMC behavior. Tier 2 estimates a prespecified four-to-six parameter subset and measures scaling with four independently initialized chains. The paper will report Tier 2 only after Tier 1 passes.

Training, validation, and posterior-region designs are disjoint in parameter vectors, shock sequences, and trajectories. Validation includes regular prior-interior points, posterior-region points, and a boundary-stress stratum. The last stratum is reported separately rather than deleted through a stability- margin screen. For stochastic extended path, evidence runs use positive branching order, multiple quadrature nodes, and nonzero coverage of every declared shock. Common case identifiers are evaluated by both the Python and MacroModelling.jl implementations.

For trajectory case $i$ with $d_x$ reported outputs, the primary approximation metric is the scale-adjusted root mean squared error

$$e_i^x = \left\{ \frac{1}{T d_x} \sum_{t=1}^{T} \left\| D_x^{-1} (\widehat{x}_{it}^F - x_{it}^F) \right\|_2^2 \right\}^{1/2}, \tag{8}$$

where $D_x$ is fixed from the training partition. Paired differences $e_{i,\mathrm{ROM}}^x - e_{i,\mathrm{sur}}^x$ are summarized with a 95 percent interval over prespecified test cases. Reporting includes medians, upper-tail quantiles, and every output separately; an average cannot hide failure in one observable.

Likelihood geometry is compared after removing an arbitrary additive constant. For reference point $\theta_0$, define

$$e^{\ell}(\theta) = \left[ \log \widetilde{L}_{\psi}(\theta) - \log \widetilde{L}_{\psi}(\theta_0) \right] - \left[ \log L_F(\theta) - \log L_F(\theta_0) \right]. \tag{9}$$

The grid analysis reports the distribution of $e^{\ell}$, rank reversals near the mode, and normalized gradient discrepancies only where the reference gradient exceeds a frozen floor. Posterior means and interval endpoints are compared to their combined grid-discretization or Monte Carlo uncertainty, not to an arbitrary number of printed decimal places.

The primary accuracy outcomes are scaled equilibrium residuals; held-out, per-output trajectory errors; paired improvement over the ROM with uncertainty; centered log-likelihood error; rank reversals near the mode; and normalized gradient error away from zero-gradient cases. Posterior comparison reports differences relative to pooled Monte Carlo error, interval endpoint differences, and a distributional diagnostic. Sampling reports rank-normalized split $\widehat{R}$, bulk and tail effective sample sizes (ESS), ESS per gradient evaluation, and ESS per second.

Timing uses the same workload, precision, and hardware for both methods. Compilation, host callbacks, device transfers, and training are reported separately. Warm likelihood microbenchmarks use repeated synchronized calls; end-to-end comparisons use independent processes. The current Schur QME path invokes a sequential SciPy host callback, so a GPU label does not imply a GPU-native likelihood. The available GPU-native matrix-sign alternative must pass Galí-specific solution and likelihood parity before it enters a speed claim.

The primary decision is sequential. Experiment A first asks whether the two-parameter nonlinear reference and surrogate conclusions agree within the frozen numerical tolerances. A failure stops escalation to the higher-dimensional sampling experiment. Conditional on correctness, Experiment B asks whether the method improves effective samples per second after offline target generation, training, compilation, transfers, and failed solves are included. A method can therefore pass the correctness gate and fail the computational-value gate.

The protocol is frozen before confirmatory execution. It records the repository revision, dirty-worktree state, environment, command, seeds, hardware, precision, and content hashes for every evidence artifact. Numerical failures remain in attempted-case denominators and are stratified by prior-interior, posterior-region, and boundary-stress designs. Machine-readable evidence is the source of manuscript tables; hand-entered numerical cells are not permitted.

## 6 Preserved engineering pilot

The pilot establishes three narrow facts. The parser, parameter-specific preflight, nonlinear equation solver, residual network, likelihood, and sampler compose on the full model. The order-zero extended-path targets satisfy a strict residual check in this small design. Finally, the Python and JAX implementations of the same surrogate profile inversion objective agree at the audited

Table 2: Preserved full-model Galí engineering pilots

| Run | Target runtime | $N_\theta \times T$ | Max. residual | Calib. RMSE | Corrected | Accept. | Draws/s |
|---|---|---|---|---|---|---|---|
| Quick smoke | fixed-reference | $4 \times 4$ | 7.21e-09 | 9.17e-05 | No | 1.000 | 0.378 |
| Corrected resource | full-jax | $4 \times 4$ | 7.21e-09 | 9.17e-05 | Yes | 1.000 | 0.0267 |
| Default pilot | fixed-reference | $8 \times 10$ | 9.92e-09 | 7.86e-06 | No | 1.000 | 8 |

*Notes:* All runs use the complete 23-equation nonlinear Galí source but estimate only two policy-rule parameters. $N_\theta$ is the number of parameter vectors and $T$ is the observation horizon. Target construction used branching order zero and one node, so these are deterministic nonlinear extended-path pilots, not stochastic-expectations SEP validation. The reported calibration RMSE is computed on the parameter block also used to fit per-output residual-shrinkage factors; it is not an untouched test error. The archived likelihood payload also duplicated equation-level innovation scales in the shock prior. "Full-JAX" recomputes the parameter-dependent steady state and first-order ROM inside the surrogate profile inversion objective. It does not evaluate a full nonlinear SEP likelihood inside HMC. Correction therefore targets the declared surrogate objective, not an exact full-nonlinear posterior. The quick, corrected-resource, and default runs use respectively 1, 2, and 1 chains with 2, 2, and 50 post-warmup draws per chain. These runs establish execution and resource feasibility only.

Table 3: First provenance-captured stochastic extended-path pilot

| Metric | Recorded value |
|---|---|
| SEP order; nodes; horizon | 1; 3; 10 |
| Complete parameter paths | 3/8 (37.5%) |
| Accepted targets | 30/80 |
| Accepted-target maximum residual | 9.41e-09 |
| Pipeline time | 119.6 minutes |
| Same-object Python/JAX difference | 3.10e-12 |
| HMC smoke: chains; warmup; retained | 1; 25; 50 |
| HMC smoke: acceptance; draws/s | 100.0%; 7.24 |
| Strict quality gate | failed |

*Notes:* Five designs, including the reference path, failed at the first simulated period. The recorded RMSE used shrinkage calibrated on the reported block, and the archived likelihood payload applied shock scales twice. The run therefore diagnoses numerical feasibility only; it is not estimation evidence.

calibration. The recorded improvement over the ROM is a calibration-set result because per-output shrinkage was fitted on the same block; it is not counted as a fourth out-of-sample fact.

The corrected resource run establishes only that endpoint evaluation is possible under a tiny budget. It retained four draws in total, achieved 100 percent acceptance, and processed 0.0267 draws per second. Those numbers are not convergence evidence and are not a speed advantage. The acceptance rate instead suggests that the static-HMC tuning is far from an efficient production regime.

The first provenance-captured positive-order run is a useful negative result. With branching order one, three quadrature nodes, and a ten-period planning horizon, only three of eight parameter designs completed all ten periods. Five designs—including the reference calibration—failed the nonlinear solve at the first simulated period. The 30 accepted targets satisfied the residual tolerance, but the strict full-path gate correctly failed. The run took 119.6 minutes, roughly 12 minutes per simulated period, on an RTX 4080 SUPER. This combination of localized first-period failures and high dense-QR cost rejects a simple "more GPU memory is sufficient" interpretation. It instead motivates continuation methods, solver-level early stopping, and a frozen correctness study before any large posterior exercise.

The clean reduced-horizon preflight provides a narrower positive result. Horizons three, four, and

Table 4: Reduced-horizon stochastic SEP evidence

| SEP horizon | Accepted paths | Maximum residual | Newton iterations |
|---|---|---|---|
| 3 | 8/8 | $1.73 \times 10^{-12}$ | 5–5 |
| 4 | 8/8 | $7.44 \times 10^{-9}$ | 4–5 |
| 5 | 8/8 | $1.93 \times 10^{-10}$ | 5–7 |
| Horizon comparison | All-state max | $\log Y$ max | $\pi^{ann}$ / $i^{ann}$ max |
| 3→4 | 0.0137 | 0.00029 | 0.000968 / 0.00112 |
| 4→5 | 0.0212 | 0.000396 | 0.00122 / 0.00139 |
| 3→5 | 0.0349 | 0.000686 | 0.00219 / 0.00251 |

*Notes:* Horizons three through five use eight prespecified path evaluations spanning seven unique points in a 12 percent perturbation design for the two policy-rule parameters; the reference calibration is evaluated twice. The calculation uses order-one stochastic extended path, three Gauss–Hermite nodes, and an $L_\infty$ residual gate of $10^{-8}$. The horizon-six hard design fails under both the sequential ($1.02 \times 10^{-4}$) and batched ($1.02 \times 10^{-4}$) solvers. First-step changes show that residual convergence is not terminal-horizon convergence. This table supports a reduced numerical pilot only; it is not likelihood or posterior evidence.

Table 5: Independent Julia–Python core parity on the complete nonlinear Galí source

| Gate | Observed | Threshold | Result |
|---|---|---|---|
| Joint design evaluations | 8/8 | 8/8 | Pass |
| Steady-state maximum absolute difference | $9.06 \times 10^{-14}$ | $10^{-10}$ | Pass |
| First-order Schur maximum absolute difference | $2.89 \times 10^{-11}$ | $10^{-8}$ | Pass |
| Python steady-state residual norm | $1.11 \times 10^{-16}$ | $10^{-10}$ | Pass |
| Clean source revisions | 2/2 | 2/2 | Pass |

*Notes:* Eight evaluations cover seven unique $(\phi_\pi, \phi_y)$ points because the reference calibration is repeated at indices 0 and 4. The local and upstream model files share SHA-256 `53f909976abc9624...`. This gate covers the non-stochastic steady state and first-order Schur solution only; nonlinear trajectory, likelihood, posterior, and GPU claims remain open.

five accept all eight path evaluations across seven unique parameter points; the maximum residual at horizon five is $1.93 \times 10^{-10}$, and the solver takes five to seven Newton iterations. Table 4 also shows why this is a pilot rather than a terminal-horizon validation. First-step outcomes continue to move as the horizon grows, and the difficult design fails at horizon six with residuals near $1.02 \times 10^{-4}$ under both solver implementations. The evidence supports stochastic target construction at a reduced horizon. It does not support a nonlinear likelihood, posterior agreement, or a computational advantage.

Table 5 establishes the first independent subgate on the complete model. Both repositories were clean and content addressed, the local and upstream Galí files had the same SHA-256 digest, and all eight evaluations passed. The worst steady-state discrepancy was $9.06 \times 10^{-14}$, while the worst first-order Schur-matrix discrepancy was $2.89 \times 10^{-11}$. These errors are well below the frozen tolerances, but the result concerns a non-stochastic steady state and local first-order solution.

Table 6 extends the independent check to nonlinear stochastic extended path at a frozen three-period horizon. Both implementations solve the complete sparse scenario tree with group counts $(1, 1, 9, 9)$ and all three first-period shock channels active. All eight evaluations pass. The worst complete-tree and expected-path discrepancies are $1.75 \times 10^{-7}$ and $7.22 \times 10^{-9}$, respectively, while both implementations remain below the $10^{-8}$ residual gate. Repeating the reference calibration at indices 0 and 4 produces identical trees within each implementation. This result closes the reduced-horizon nonlinear-tree subgate only. It neither resolves the observed horizon-five-to-six

Table 6: Independent Julia–Python nonlinear SEP parity on the complete Galí source

| Gate | Observed | Threshold | Result |
|---|---|---|---|
| Joint design evaluations | 8/8 | 8/8 | Pass |
| Complete scenario-tree maximum absolute difference | $1.75 \times 10^{-7}$ | $10^{-6}$ | Pass |
| Expected-path maximum absolute difference | $7.22 \times 10^{-9}$ | $10^{-8}$ | Pass |
| Python solver residual norm | $9.23 \times 10^{-9}$ | $10^{-8}$ | Pass |
| Julia solver residual norm | $9.21 \times 10^{-9}$ | $10^{-8}$ | Pass |
| Clean source revisions | 2/2 | 2/2 | Pass |

*Notes:* The test uses a three-period, order-one sparse tree with three Gauss–Hermite nodes per shock, group counts $(1, 1, 9, 9)$, and first-period standardized innovations $(0.12, -0.08, 0.10)$ across all three shock channels. Eight evaluations cover seven unique $(\phi_\pi, \phi_y)$ points; indices 0 and 4 repeat the reference calibration and reproduce bit-for-bit within each implementation. This closes only the reduced-horizon nonlinear-tree subgate; production-horizon likelihood, posterior, and GPU gates remain open.

instability nor validates a nonlinear likelihood or posterior.

# 7 Claim-specific decision rules

Table 7 defines the minimum evidence needed to promote the working draft to a submission. Counts are protocol defaults, not universal publication standards; learning curves, uncertainty intervals, and reference convergence determine whether they are sufficient.

All thresholds and case identifiers will be frozen before the final runs. The evidence builder reads raw result files, verifies the full-model identity, hashes its sources, and emits both machine-readable and manuscript inputs. A failed hard gate remains visible in the replication package; it is not converted into a missing observation or silently omitted path.

# 8 Limitations and interpretation

The preserved evidence is an engineering pilot. The earlier nonlinear targets use branching order zero and one node, so they do not validate stochastic expectations. The later order-one, three-node evidence establishes target feasibility only through a five-period terminal horizon; first-step sensitivity and the horizon-six failure prevent a terminal-convergence claim. Most likelihood calls hold the reference steady state and ROM fixed. The reported surrogate RMSE values are calibration-adjusted rather than measured on an untouched test partition. The parameter-dependent full-JAX run still targets the surrogate profile inversion objective, assigns a finite $-10^{12}$ sentinel to failed solves, and uses a host-callback Schur solve. Its archived payload also duplicated the equation-level innovation scales in the inversion prior; future evidence uses unit-variance standardized innovations. An exact structural posterior instead requires explicit support handling and an economically defined likelihood. The sampler budgets are too small for convergence diagnostics, and the data are synthetic and short. The independent nonlinear Julia comparison is restricted to a three-period sparse tree, a single frozen first-period shock vector, and seven unique policy-rule points. It does not establish production-horizon trajectory or likelihood parity. The current evidence therefore supports neither exact nonlinear posterior sampling nor an end-to-end speedup claim.

These limitations determine the paper's next experiments rather than merely qualifying its conclusion. A credible methods result requires the two-parameter reference experiment to pass before scaling the posterior dimension. If the surrogate does not preserve the grid posterior, or if its

Table 7: Hard gates for a credible reduced-model methods submission

| Evidence family | Required result |
|---|---|
| Scope and provenance | Full 23-equation source verified by content hash and upstream revision; frozen synthetic-data generator and estimand; clean repository revision; complete environment, data, command, seed, and artifact manifests. |
| Nonlinear solution | Positive-order stochastic SEP with multiple nodes and all shocks active; raw attempted and successful counts; scaled residuals; regular, posterior, and boundary-stress strata. |
| Independent parity | Common case identifiers evaluated in Python and MacroModelling.jl; both failures retained in the denominator; QZ, trajectory, and likelihood errors reported on common successful cases. |
| Surrogate accuracy | Disjoint parameter, shock, and trajectory partitions; shrinkage disabled or fitted on a separate calibration partition; untouched-test per-output and tail errors; paired improvement over ROM with a 95 percent interval; learning-curve and out-of-distribution results. |
| Posterior fidelity | Two-parameter full nonlinear grid with resolution-doubling convergence; posterior moments, intervals, and distribution compared with the surrogate and corrected samplers using numerical or Monte Carlo uncertainty. |
| Sampling | Four tuned chains for the scaling experiment; rank-normalized split $\widehat{R}$, bulk and tail ESS, failures, ESS per gradient evaluation, and ESS per second. |
| Timing | Same-work, same-precision comparison; synchronized warm calls and independent end-to-end processes; compilation, transfers, training, host callbacks, and peak memory reported separately. |

*Notes:* These are claim-specific design requirements. Prespecified thresholds and case lists must be frozen before confirmatory execution. Passing them supports the reduced Galí experiment only; it does not establish universal accuracy across DSGE models.

full accounting does not improve ESS per second, the negative result should be reported and the claim reduced to numerical infrastructure.

# 9 Conclusion

Residual learning may be useful for nonlinear DSGE estimation because a first-order model provides a disciplined baseline and the network concentrates on the nonlinear discrepancy. Target-corrected HMC can remove error from the trajectory force, but only relative to the density evaluated at the Metropolis endpoints. The Galí evidence now establishes order-one stochastic target construction across eight pilot path evaluations and seven unique parameter points at a five-period horizon. It also shows why that result is insufficient: outcomes still move with the terminal horizon, and the hard design fails at horizon six. Publication readiness therefore depends on terminal-horizon validation, an untouched test design, an independently converged nonlinear posterior reference, production-horizon likelihood parity, efficient multichain sampling, and same-work timing. The steady-state, first-order, and reduced three-period nonlinear-tree portions of the cross-language gate now pass on the full source. The protocol makes the remaining requirements falsifiable and preserves negative results rather than redefining success after a run.

## Disclosure on computational assistance

Automated coding and language tools were used to inspect software, generate tests, and edit the working draft. Every numerical claim in the paper is linked to a machine-readable source file and must be verified by the authors. The final submission will follow the target journal's disclosure policy.

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